CIVL1105 Environmental Engineering

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The teaching team



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Little about myself

- PhD (2019) in Wastewater Engineering from Advanced Water Management Centre, The University of Queensland
- Senior Lecturer (2024) in School of Civil and Environmental Engineering, The University of New South Wales, Sydney





A little bit about you (Mentimeter)

Introduction to Environmental Engineering and Science

Lecturers

Dr. Haoran DUAN & Dr. Lu Elfa PENG,

US National Academy of Engineering

Greatest Engineering Achievements of the 20th Century

◆ About ◆ Timeline ◆ The Book

Welcome!

How many of the 20th century's greatest engineering achievements will you use today? A car? Computer? Telephone? Explore our list of the top 20 achievements and learn how engineering shaped a century and changed the world.

1. Electrification
2. Automobile
3. Airplane
4. Water Supply and Distribution
5. Electronics
6. Radio and Television
7. Agricultural Mechanization
8. Computers
9. Telephone
10. Air Conditioning and Refrigeration
11. Highways
12. Spacecraft
13. Internet
14. Imaging
15. Household Appliances
16. Health Technologies
17. Petroleum and Petrochemical Technologies
18. Laser and Fiber Optics
19. Nuclear Technologies
20. High-performance Materials



Greatest Achievements

Greatest Engineering Achievements of the 20th Century

Water Supply and Distribution

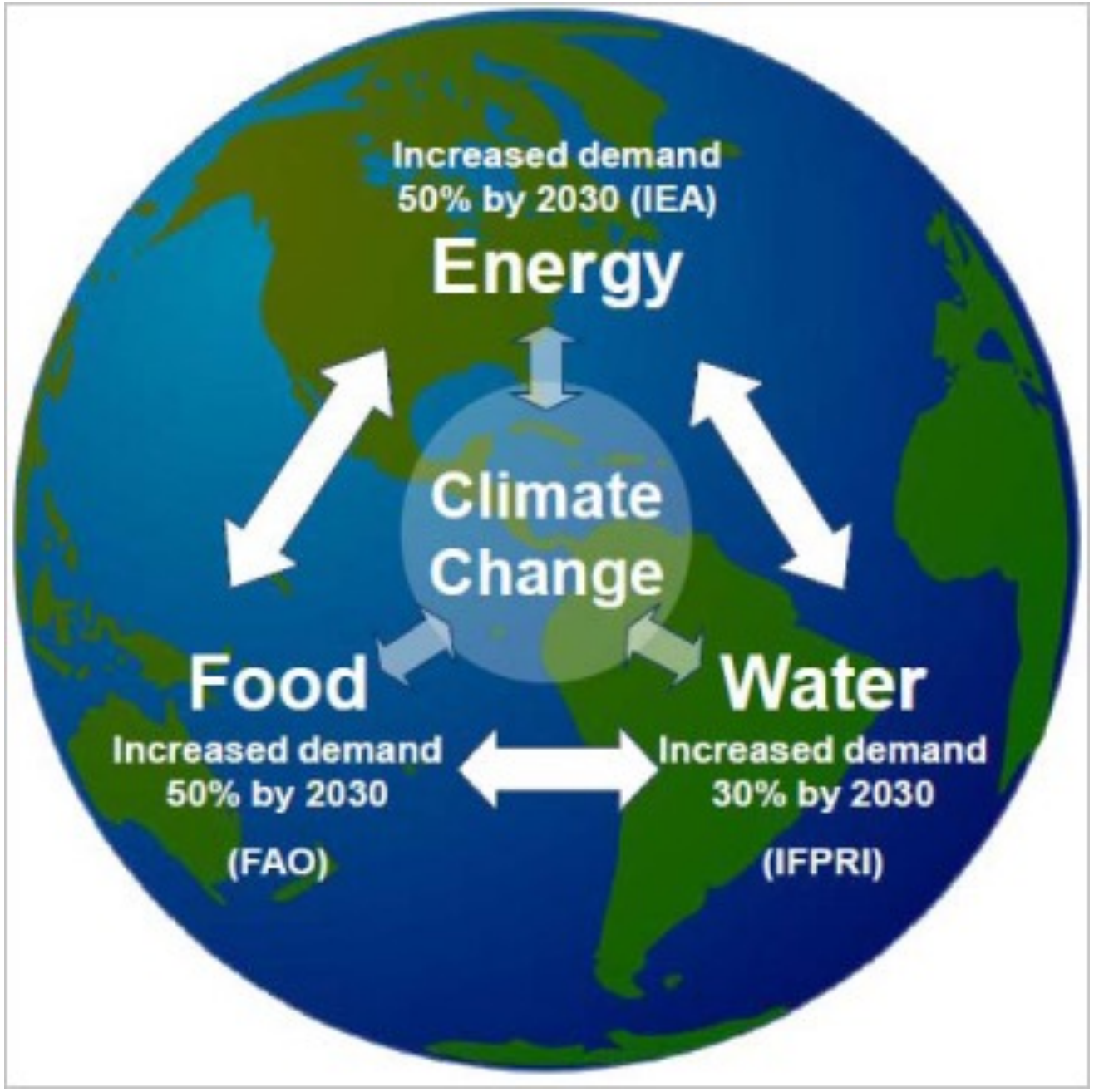
"At the beginning of the 20th century, in the United States and in many other countries, water was both greatly in demand and greatly feared... At the same time, the condition of existing water supply systems was abysmal—and a direct threat to public health."

- Waterborne diseases (e.g., diarrhea) were the 3rd leading cause of death.
- Cholera, typhoid fever …

http://www.greatachievements.org/

Grand challenges of our time!

- Energy
- Water
- Food



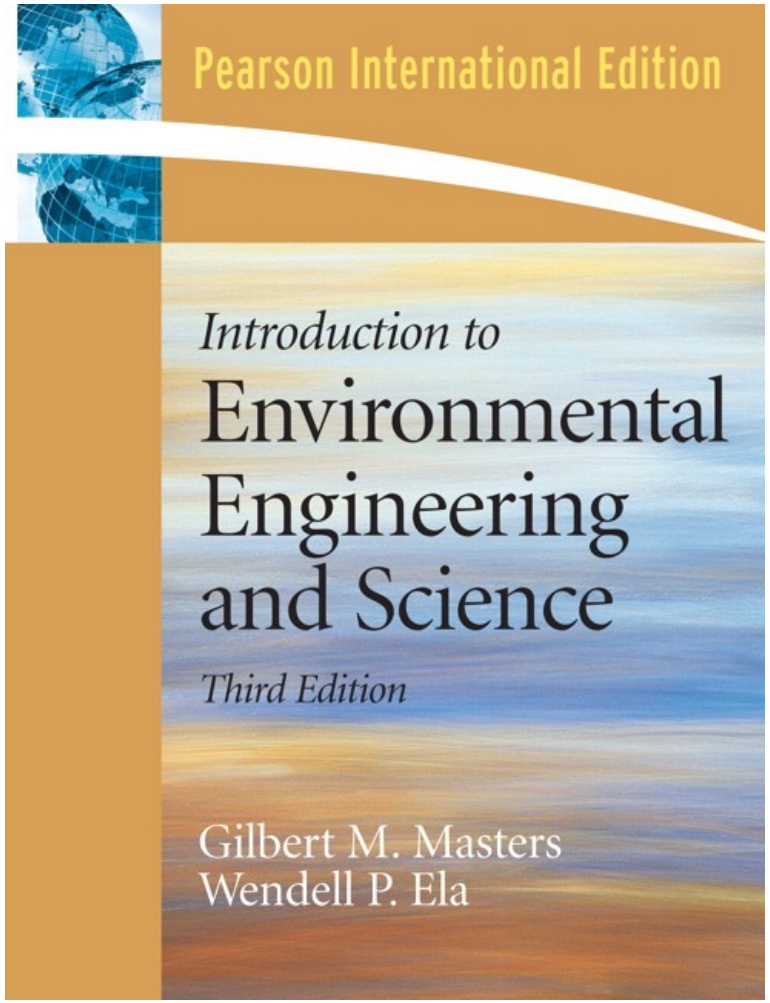
http://www.waternexussolutions.org/284/the-perfect-storm-scenario-and-nexus-thinking-to-meet-increasing-demand-for-energy-food-water.html#.V8ZWyvl9670

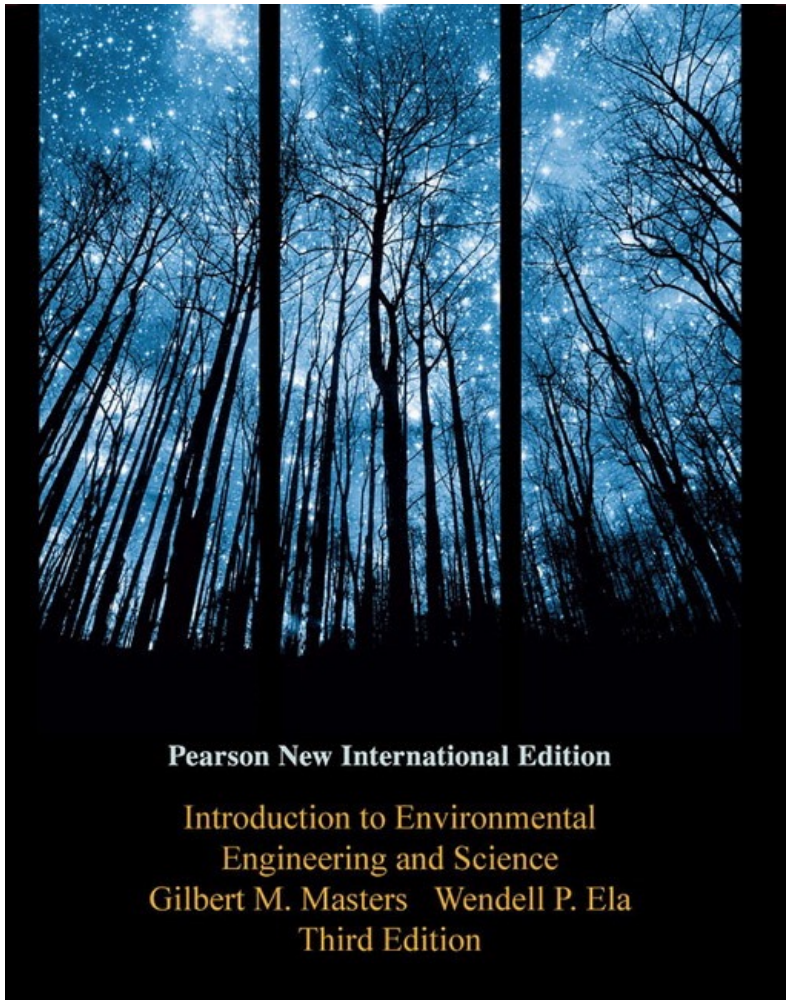
Environmental Engineering and Science

- A **multidisciplinary** academic field (physics, chemistry, biology, …)
- Focusing on the **interactions** of living (particularly humans) and non-living components of the environment
- providing **solutions** to environmental problems
- Examples of environmental issues:
  - Pollution (water pollution, air pollution, noise pollution, …)
  - Wastes
  - Global atmospheric change (global warming and climate change, ozone depletion)
  - Environmental degradation
  - Environmental health
  - …

Textbook

Introduction to Environmental Engineering and Science: International Edition (3e) by Gilbert Masters and Wendell Ela





Topics and Schedule

Week	Tuesday (9:00-9:50) MB237	Friday (13:00 – 14:50) MB237
1	Introduction to Environmental Engineering	Mass Balance
2	Tutorial (Yabo)	Energy Fundamentals (Dr. Peng)
3	Environmental Chemistry (Dr. Peng)	Environmental Chemistry (Dr. Peng)
4	Tutorial (Zhiheng + Bin)	Water Pollution
5	Water Pollution	Drinking Water Treatment
6	Drinking Water Treatment	Wastewater Treatment Overview and Preliminary/primary Treatment
7	Reading week	
8	Biological Wastewater Treatment Principals	Cancelled*
9	Biological Wastewater Treatment Principals	Activated Sludge Process
10	Activated Sludge Process & BNR	Biological nutrient removal
11	Tutorial (Yuantian)	Solid waste management and resource recovery
12	Solid waste management and resource recovery	Air Pollution
13	Air Pollution	Climate Change & Course final review

*As approved by the Senate, **all UG and TPG classes scheduled on October 23, 2026 [Fri] afternoon (starting from 1:00 pm onwards) and October 24, 2026 [Sat] will be CANCELLED to facilitate the necessary preparations for the Information Day events**. Hence, no classes will be scheduled.

Topics and Schedule

Teaching Weeks 2026-27 for Undergraduate and Taught Postgraduate Students

	SUN	MON	TUE	WED	THUR	FRI	SAT	FIRST SEMESTER: SEP 1 - DEC 23, 2026	Week
SEP-26			1	2	3	4	5	First Day of Teaching: Sep 1, 2026	1
	6	7	8	9	10	11	12		2
	13	14	15	16	17	18	19		3
	20	21	22	23	24	25	[26]		4
	27	28	29	30					5
OCT-26					[1]	2	3		
	4	5	6	7	8	9	10		6
	11	12	13	14	15	16	17	Reading/Field Trip Week: Oct 12 - 17, 2026	7(Reading)
	18	[19]	20	21	22	23	24		8
	25	26	27	28	29	30	31		9
NOV-26	1	2	3	4	5	6	7		10
	8	9	10	11	12	13	14		11
	15	16	17	18	19	20	21		12
	22	23	24	25	26	27	28		13
	29	30						Last Day of Teaching: Nov 30, 2026	
DEC-26			1	2	3	4	5	Revision Period: Dec 1 - 5, 2026	14(Revision)
	6	7	8	9	10	11	12	Assessment Period: Dec 7 - 23, 2026	1
	13	14	15	16	17	18	19		2
	20	21	22	23	(24)	[25]	[26]		3
	27	28	29	30	<31>				Break

Note:
Face to face delivery, no recording.

No class scheduled in reading week.

No class on 23rd Oct as per university arrangement

Grading

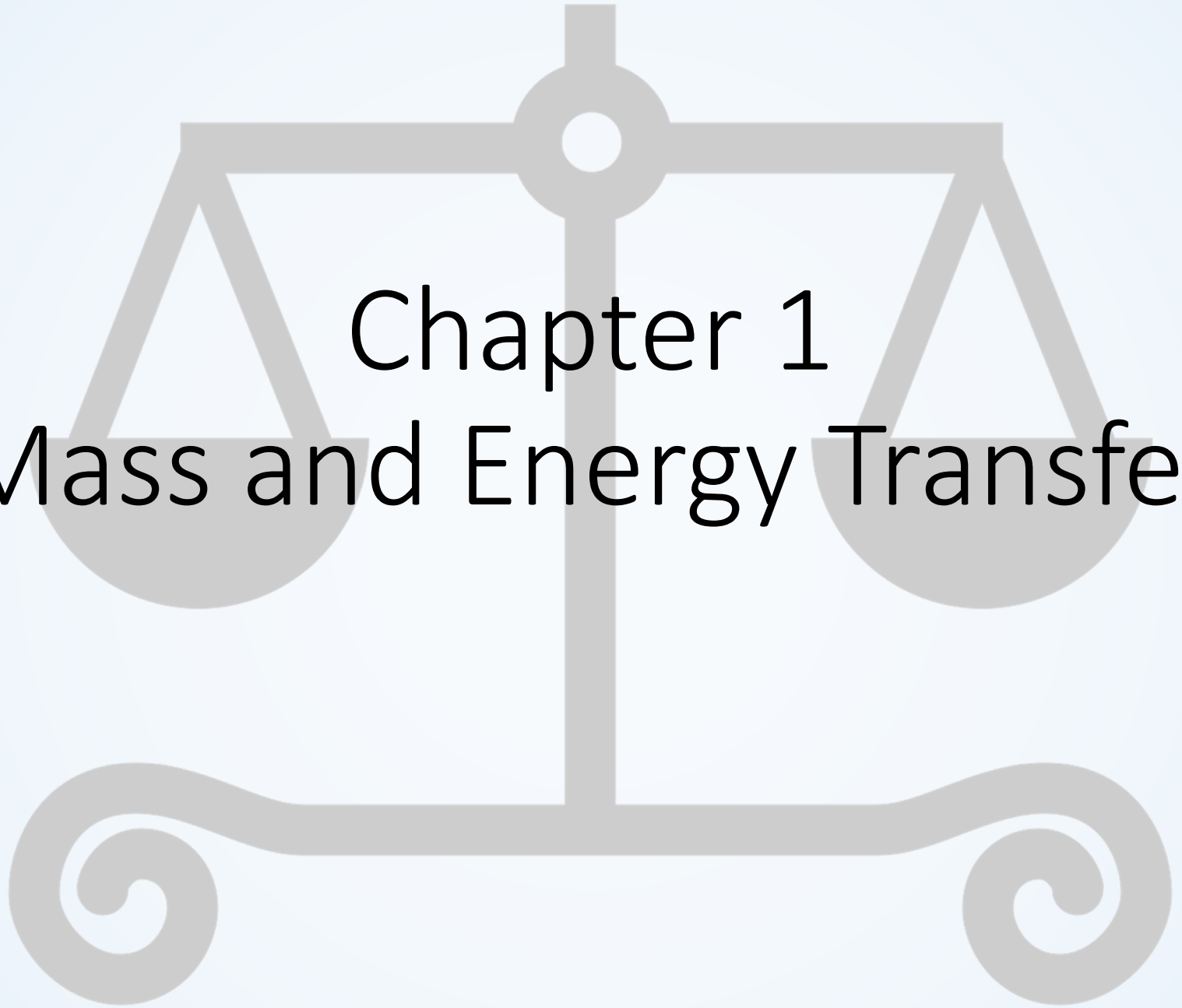
A+	A	A-	B+	B	B-
8	4	9	4	5	6
26.3%			18.8%		

A+	A	A-	B+	B	B-
2	3	5	4	4	10
13.2%			23.7%		

- Exam (**80%**)

- Assignments (**20%**)
  - Details to be announced later

- Additional practice problems (not graded)
  - Suggested practice problems
  - Solution will be provided



Chapter 1
Mass and Energy Transfer

Expected learning outcomes for Chapter 1

- Basics
  - Unit conversion
  - **Ideal gas law**
- **Mass balance and reactor model**
  - Reactor rate (zeroth-order, **1st order**, and 2nd order)
  - **Mass balance**
  - **Reactor model (CSTR vs. plug flow reactor)**
- **Energy**
  - **Laws of thermodynamics** (0th, **1st, and 2nd**)
  - Heat, specific heat, latent heat, and phase change

Engineering unit conversions

Environmental engineering decisions depend on measurement.

Examples:

- Designing a water treatment plant (ML/day → m^3/s)
- Estimating pollutant loads (mg/L × ML/day → kg/day)
- Calculating greenhouse gas emissions (kg CO_2/day)
- Air pollution concentrations (ppm ↔ mg/m^3)

A number without a unit is usually not useful.

Commonly used units

Quantity	Unit	Common Engineering Units
Length	m	mm, cm, km
Area	m^2	m^2
Volume	m^3	L, mL, ML
Mass	kg	mg, g, tonne
Time	s	min, h, day
Concentration	mg/L	g/L, kg/m^3
Flow	m^3/s	L/s, ML/day

Note:

Volume ≠ flow

A reservoir may contain a volume of 10^6 m^3, while the flow leaving it may be 2 m^3/s.

mass ≠ mass load

A tank may contain 500 kg of a substance, while 50 kg/day enters the tank.

Unit prefixes

Commonly used unit prefixes

Prefix	Symbol	Multiplication factor	Example
mega	M	10^6	Megalitre (ML) = 1×10^6 L
kilo	k	10^3	Kilometre (km) = 1×10^3 m
milli	m	10^{-3}	Millilitre (mL) = 1×10^{-3} L
micro	µ	10^{-6}	Microgram (µg) = 1×10^{-6} g

Quantity	Prefix	Example
10^{-9}	nano	**ng; nm**
10^9	giga	**GW**
10^{12}	tera	**TW**

Unit conversion

Mass conversion

- 1,000,000 µg (10^6) = 1,000 mg (10^3) = 1 g = 0.001 kg (10^{-3}) = 10^{-6} tonne

Length conversion

- 1,000,000 µm (10^6) = 1,000 mm (10^3) = 1 m = 0.001 km (10^{-3})

Volume conversion

- 1,000,000 µL (10^6) = 1,000 mL (10^3) = 1 L = 0.001 m^3 (10^{-3})
- 1 ML = 1,000 m^3
- 1 GL = 1000 ML = 1,000,000 m^3

Three quantities describe a discharge

1. Flow: How much water or air moves?

$$Q = \frac{\text{volume}}{\text{time}}$$

Typical units: L/s; m^3/s; ML/d.

2. Concentration: How strong is the pollution?

$$C = \frac{\text{mass of pollutant}}{\text{volume}}$$

Typical units: mg/L for water; $\mu g/m^3$ for air.

3. Mass load: How much pollutant is transported?

$$L = CQ$$

Typical units: kg/d; tonnes/year.

Units of Concentration

$$C = \frac{\text{Mass of pollutant}}{\text{Volume}}$$

Mass: µg, mg, g, kg, …

Volume: mL, L, m^3, …

Typical units: mg/L for water; µg/m^3 for air.

Question

A river sample has a nitrate concentration of 5 mg/L. What does this mean in words?

It means there are 5 mg of nitrate in every 1 L of water. (note this is different from 5mgN/L)

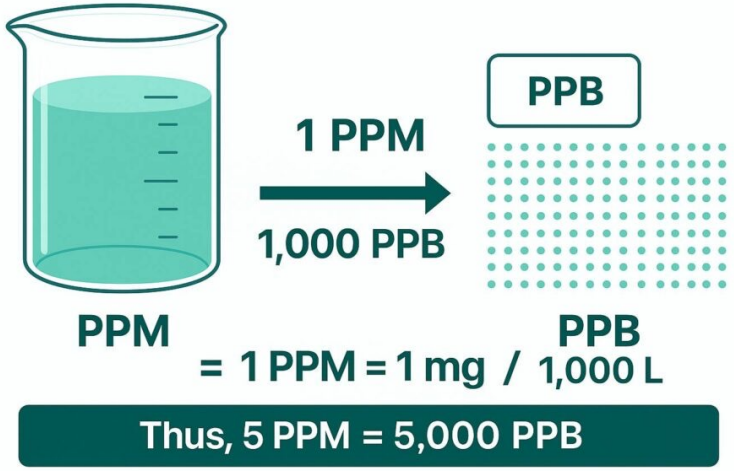
Units of Concentration

For dilute aqueous solutions with **density close to 1 kg/L**:

- 1 mg/L = 1 ppm (parts per million)
- 1 µg/L = 1 ppb (parts per billion)
- 1 mg/L = 1,000 µg/L
- 1 ppm = 1,000 ppb

For dilute water:

- 1 mg/L = 1 g/m^3 = 0.001 g/L = 0.001 kg/m^3



Units of Concentration

- Convert 1 µg/L to 1 mg/L

  Given 1,000 µg = 1 mg

$$\text{Concentration in mg/L} = \frac{\text{Concentration in µg/L}}{1000}$$

- Convert 1 mg/L to 1 kg/m^3

  Given 1,000,000 mg = 1 kg & 1,000 L = 1 m^3

$$\frac{1\ \text{mg}}{\text{L}} \times \frac{1\ \text{kg}}{1{,}000{,}000\ \text{mg}} \times \frac{1{,}000\ \text{L}}{1\ \text{m}^3} = \frac{1{,}000}{1{,}000{,}000}\text{kg/m}^3 = 0.001\text{kg/m}^3$$

Concentration units (liquid)

- Mass concentration
  - mg/L; g/L
  - g/m^3
  - ppm (parts per million); ppb (parts per billion)
  - For water:
    - 1 mg/L → 1 mg of solutes per kg of water → 10^{-6} → 1 ppm
    - 1 mg/L = 1 g/m^3 = 1 ppm (by weight)
    - 1 μg/L = 1 mg/m^3 = 1 ppb (by weight)
  - Other liquids
    - mg/L = ppm (by weight) x specific gravity of mixture
- Molar concentration
  - mol/L or M

    (Avogadro's number: 6.02 x 10^{23} molecules/mol)

1. When to prepare 5 ppm $NaHCO_3$ in 1 L of water solution, how much $NaHCO_3$ is needed?
(A) 5 mg; (B) 5 g; (C) 84 mg; (D) 84 g.

Read example 1.1

Gases

- Common units
  - mg/L; μg/L
  - ppmv (or simply ppm)
    - 1 ppmv = 1 ppm (by volume) = 1 volume of gaseous pollutant/10^6 volumes of air

All the gasses (e.g., O_2, CO_2, and air) in this course can be considered as the ideal gas

- Ideal gas law

  PV = nRT

  Where

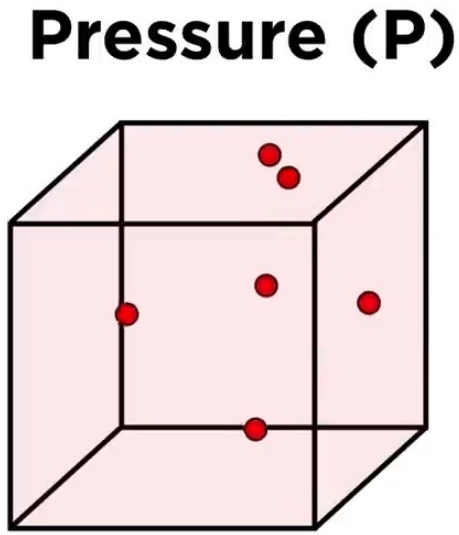
  P = absolute pressure (Pa)

  V = volume (m^3)

  n = amount of substance (mol)

  R = ideal gas constant (**8.31 J/(K·mol)**

  T = absolute temperature (K); **K = °C + 273.15**



Gases

When the volume ratio between two ideal gases (v1/v2) is 2/3, what is their molar ratio (n1/n2)? (A) 2/3; (B) 3/2; (C) 8.31; (D) 22.4.

- Common units
  - mg/L; μg/L
  - ppmv (or simply ppm)
    - 1 ppmv = 1 ppm (by volume) = 1 volume of gaseous pollutant/10^6 volumes of air

All the gasses (e.g., O_2, CO_2, and air) in this course can be considered as the ideal gas

- Ideal gas law

  PV = nRT

  Where

  P = absolute pressure (Pa)

  V = volume (m^3)

  n = amount of substance (mol)

  R = ideal gas constant (**8.31 J/(K·mol)** or 0.082056 L·atm/(K·mol))

  T = absolute temperature (K); **K = °C + 273.15**

$$\frac{V_1}{n_1} = \frac{RT}{P} = \frac{V_2}{n_2} \rightarrow \frac{V_1}{V_2} = \frac{n_1}{n_2}$$

$$J = N \cdot m = N/m^2 \times m^3$$

@S.T.P = standard temperature (0°C)
standard pressure (1atm = 1.01×10^5 Pa)

Mass load

$$\text{Mass load} = \text{concentration} \times \text{flow}$$

Convenient load calculation for water systems

$$L\ (\text{kg/d}) = C\ (\text{mg/L}) \times Q\ (\text{ML/d})$$

This works because:

$$Mass\ load\ \left(\frac{kg}{d}\right) = C\left(\frac{mg}{L}\right) \times Q\left(\frac{m^3}{d}\right) \times \frac{10^{-6}\ kg}{1\ mg} \times \frac{1\ L}{10^{-3} m^3}$$

Hence

$$1\ \text{mg/L} \times 1\ \text{ML/d} = 1\ \text{kg/d}$$

High concentration does not always mean high load

Discharge	Concentration	Flow	Load
A: Small industrial stream	100 mg/L	1 ML/d	? kg/d
B: Large municipal stream	20 mg/L	10 ML/d	? kg/d

High concentration does not always mean high load

Discharge	Concentration	Flow	Load
A: Small industrial stream	100 mg/L	1 ML/d	100 kg/d
B: Large municipal stream	20 mg/L	10 ML/d	200 kg/d

- Discharge A has the higher concentration.
- Discharge B has the higher total pollutant load.
- Concentration is important for toxicity and compliance.
- Load is important for treatment capacity and total environmental input.

Gases

Ex. 1.2: Find the volume that 1 mole of an ideal gas would occupy at standard temperature and pressure (STP) conditions of 1 atmosphere of pressure and 0 °C temperature. Repeat the calculation for 1 atm and 25 °C.

Solution: Using ***PV=nRT*** at a temperature of 0 °C (273.15 K) gives

$$V = \frac{1\ \text{mol} \times 0.082056\ \text{L} \cdot \text{atm} \cdot \text{K}^{-1} \cdot \text{mol}^{-1} \times 273.15\ \text{K}}{1\ \text{atm}} = \mathbf{22.414\ L}$$

(@STP)

and at 25 °C (298.15 K)

$$V = \frac{1\ \text{mol} \times 0.082056\ \text{L} \cdot \text{atm} \cdot \text{K}^{-1} \cdot \text{mol}^{-1} \times 298.15\ \text{K}}{1\ \text{atm}} = \mathbf{24.465\ L}$$

(@SP, T=25°C)

Appendix A. Information for Reference

I. Atomic weights (g/mol)

H 1; C 12; N 14; O 16; F 19; Na 23; Mg 24.3; Al 27; P 31; S 32; Cl 35.5; K39; Ca 40; Cr 52; Mn 55; Fe 55.8; Ni 58.7; Cu 63.5; Zn 65.4; I 127.

Molecular weights (g/mol)

For example: CO = 12 + 16 = 28

O_2 = 16 × 2 = 32

CO_2 = 12 + 16 × 2 = 44

II. Properties of water

Density = 1000 kg/m^3

Viscosity (15 ˚C) = 1.1404 mPa·s

Specific heat = 4.18 kJ/(kg·˚C)

Heat of vaporization (15 ˚C) = 2465 kJ/kg

Heat of fusion = 333 kJ/kg

III. Other constants

Ideal gas law constant = 0.082056 L·atm/(K·mol)

Henry's Law constant for CO_2: 0.0334 mol/(L·atm)

Acidity constants for H_2CO_3: pK_1 = 6.3; pK_2 = 10.3

Solubility of $CaCO_3$(s): pK_{so} = 8.3

Henry's Law constant for O_2 (25 ˚C) = 0.00126 mol/(L·atm)

Dissociation constant of water (25 ˚C) $K_w = 1\times10^{-14}$

Equivalent Weight (EW) for Ca^{2+} = 20 mg/meq

EW for Mg^{2+} = 12.2 mg/meq

EW for $CaCO_3$ = 50 mg/meq

Concentration units (Gas)

Ex. 1.3: U.S. Air Quality Standard for carbon monoxide (CO) is 9 ppmv. Express this standard in % as well as in mg/m^3 (at 1 atm and 25 °C)

$\frac{V}{n} = 24.5\ L/mol$

Solution : % CO = 9 x 10^{-6} x 100% = 0.0009%

Concentration units (Gas)

Ex. 1.3: U.S. Air Quality Standard for carbon monoxide (CO) is 9 ppmv. Express this standard in % as well as in mg/m^3 (at 1 atm and 25 °C)

Solution : % CO = 9 x 10^{-6} x 100% = 0.0009%

$\frac{V}{n} = 24.5\ L/mol$

mg/m^3

① 9 ppm = $\frac{9\times10^{-6}\ m^3\ CO}{1\ m^3\ air}$ = $9\times10^{-3}\ L\ CO$

Volume/Volume

$= \frac{9\times10^{-3}\ L}{24.5\ L/mol} = 0.37\ mol\ CO$

$= 0.37\ mol \times 28\ g/mol = 10.3\ mg\ CO/m^3\ air$

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② 9 ppm = $\frac{9\times10^{-6}\ mol\ CO}{1\ mol\ air}$ = $9\times10^{-6}\ mol \times 28\ g/mol = 0.252\ mg\ CO$

mol/mol

$= 1\ mol \times 24.5\ L/mol = 24.5\times10^{-3}\ m^3\ air$

$= \frac{0.252\ mg\ CO}{24.5\times10^{-3}\ m^3\ air} = 10.3\ mg/m^3\ \frac{CO}{air}$

Past paper 2025-S1 Single-choice Q1

CO_2 in the atmosphere has reached a record high of 427 ppm this year. What is the CO_2 concentration in the unit of g/m^3 at standard pressure and room temperature? Given MW of CO_2 = 44 g/mol.

$$427\ \text{ppm} = \frac{427\times10^{-6}\ m^3\ of\ CO_2}{m^3\ of\ air}$$

$$427\times10^{-6}\ m^3 = 427\times10^{-3}\ \text{L}$$

$$\frac{0.427\ \text{L}}{24.465\ \text{L/mol}} = 0.0175\ \text{mol}$$

$$0.0175\ \text{mol}\times 44\ \text{g/mol} = 0.77\ \text{g} \rightarrow 427\ \text{ppm} = 0.77\ \text{g/m}^3$$

Mass balance

Environmental-engineering calculations are based on the principle of mass balance.

$$\text{Accumulation} = \text{Input} - \text{Output} + \text{Generation} - \text{Consumption}$$

- **Accumulation:** Change in mass stored in the system
- **Input:** Mass entering the system
- **Output:** Mass leaving the system
- **Generation:** Mass produced by physical, chemical or biological processes
- **Consumption:** Mass removed or transformed by reactions

Pollutants do not disappear. They leave the system, accumulate, or are transformed into other substances.

Mass balance

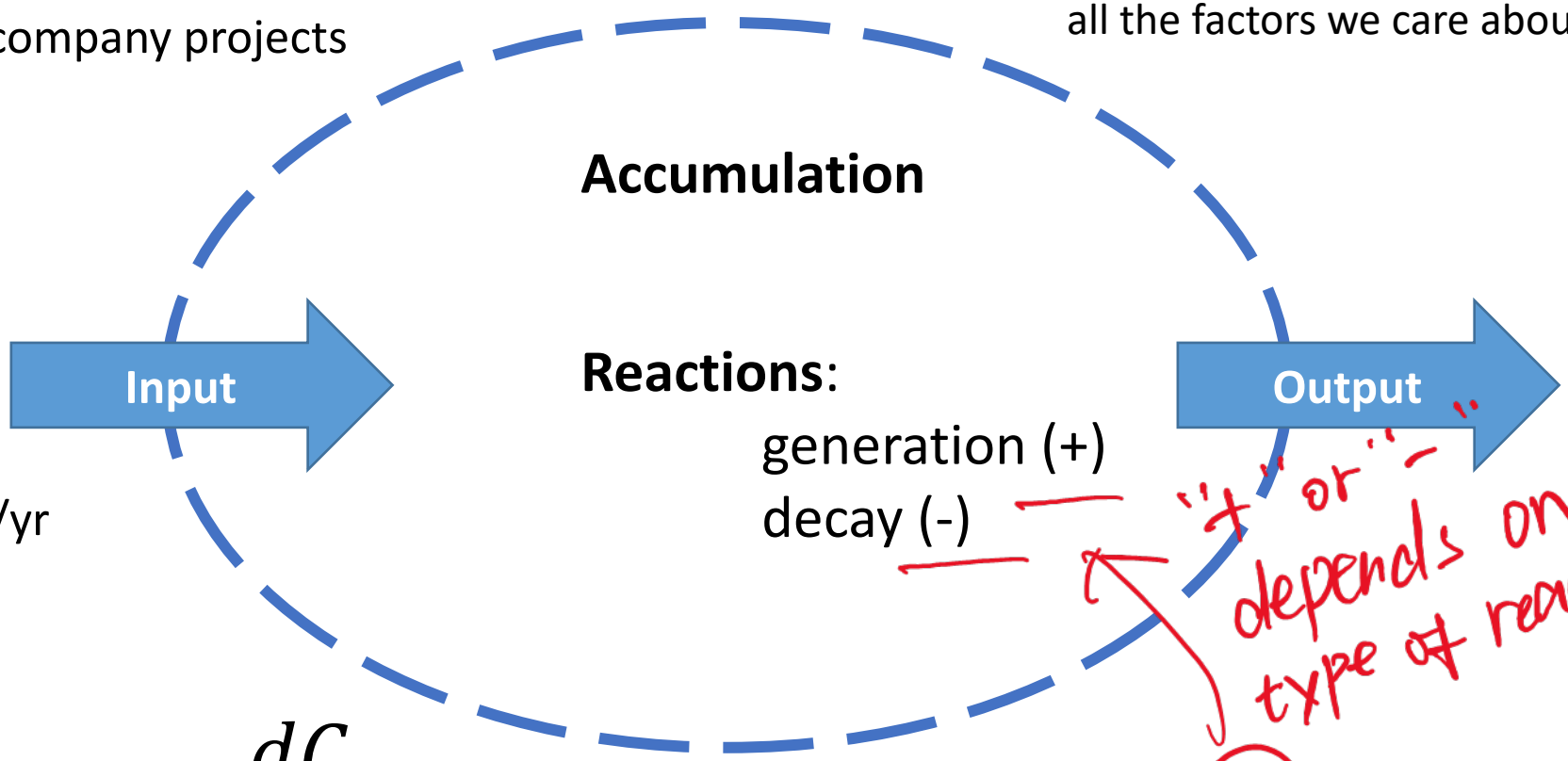
A rough example

System: a bank account for company projects (control volume)
Mass: money
Time: year

Input rate: savings/yr
Output rate: expense/yr
Reaction rate:
generation: bank interests/yr
decay: bank charge/yr
Accumulation rate:
bank account balance/yr

Input flow rate (Qin):
number of projects/yr
Input concentration (Cin):
single project income/yr

Control volume: a container comprising all the factors we care about



$$V\frac{dC}{dt} = QinCin - Q_{out}C_{out} + Vr(c)$$

Accumulation rate = **Input rate** - **Output rate** + **Reaction rate**

Mass balance

$$V\frac{dC}{dt} = QinCin - Q_{out}C_{out} + VkC^n$$

Accumulation rate = Input rate - Output rate + Reaction rate

Steady state → accumulation rate = 0
Conserved substance → reaction rate = 0

Conserved substance: total dissolved solids, heavy metals in soil, CO_2
Non-conserved substance: organic matter (decay), biomass (growth)

For a treatment process:

$$M_{\text{in}} = M_{\text{out}} + M_{\text{removed}}$$

Removal efficiency is:

$$\eta = \frac{M_{\text{in}} - M_{\text{out}}}{M_{\text{in}}} \times 100\%$$

Mass balance

– Steady state conservative systems

In 1 L of volume

C_{solute} = ? mg/L

C_{water} = 1 kg/L ρ - ? mg/L

$C_1 \approx C_2 \approx C_3$

Accumulation rate = **Input rate** - **Output rate** + **Reaction rate**

$0 = Input_1 + Input_2 - Output$

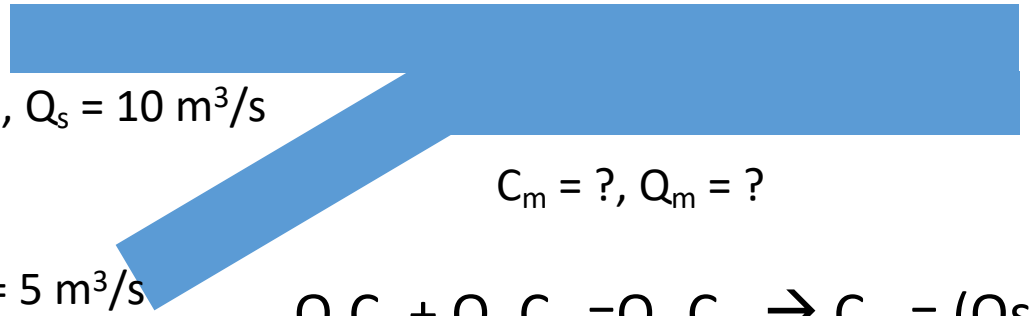
Input rate = Output rate

$0 = Q_s C_s + Q_w C_w - Q_m C_m$

$\rho Q_s + \rho Q_w = \rho Q_m$

$C_m = \frac{Q_s C_s + Q_w C_w}{Q_s + Q_w}$

Ex. 1.4



C_s = 20 mg/l, Q_s = 10 m^3/s

C_m = ?, Q_m = ?

C_w = 40 mg/l, Q_w = 5 m^3/s

$Q_s C_s + Q_w C_w = Q_m C_m \rightarrow C_m = (Q_s C_s + Q_w C_w)/Q_m$

$\rho Q_m = \rho Q_s + \rho Q_w$ = 10 + 5 = 15 m^3/s (by mass balance of water)

or

$Q_m = Q_s + Q_w$ = 15 m^3/s (by assuming a fixed control volume)

C_m = (20 x 10 + 40 x 5) / 15 = 26.7 mg/l

Mass balance

A wastewater treatment plant receives a COD load of 4500 kg/d. The plant removes 80% of the COD. Calculate COD loads leaving the plant.

Mass balance: practice

A wastewater treatment plant receives a COD load of 4500 kg/d. The plant removes 80% of the COD. Calculate COD loads leaving the plant.

Solution:

Removed load

$$M_{\text{removed}} = 80\% \times 4{,}500\text{kg/d} = 3{,}600\ \text{kg/d}$$

Effluent load

$$M_{\text{out}} = M_{\text{in}} - M_{\text{removed}} = 4{,}500 - 3{,}600 = 900\ \text{kg/d}$$

Mass-balance check

$$4{,}500 = 900 + 3{,}600$$

Mass balance

– Batch systems with nonconservative pollutants

Accumulation rate = ~~**Input rate**~~ - ~~**Output rate**~~ + **Reaction rate** $V\frac{dC}{dt} = VkC^n$

Accumulation rate = Reaction rate

differential equation

$V\frac{dC}{dt} = Vr(C) = VkC^n$

$\frac{dC}{dt} = r(C) = \text{reaction rate}$

integral equation

$V\int_{C_0}^{C} dC = V\int_0^t kC^n dt$

$\int_{C_0}^{C} dC = \int_0^t k\,dt$

k (zero-order) → **$C = C_0 + kt$**

0th order reaction: mass change over time is **not dependent** on the amount of the substance present. e.g., water evaporation from a cup,

$\int_{C_0}^{C} dC = \int_0^t kC\,dt$

kC (first-order) → **$C = C_0\, e^{kt}$**

1st order reaction: the mass change over time **is directly proportional** to the mass that is originally present. e.g., bacteria growth, population dynamics

kC^2 (second-order)

How to define the order of your reaction?

Mass balance

– Steady systems with nonconservative pollutants

$\frac{dC}{dt} = 0$ ~~Accumulation rate~~ = **Input rate** - **Output rate** + **Reaction rate**

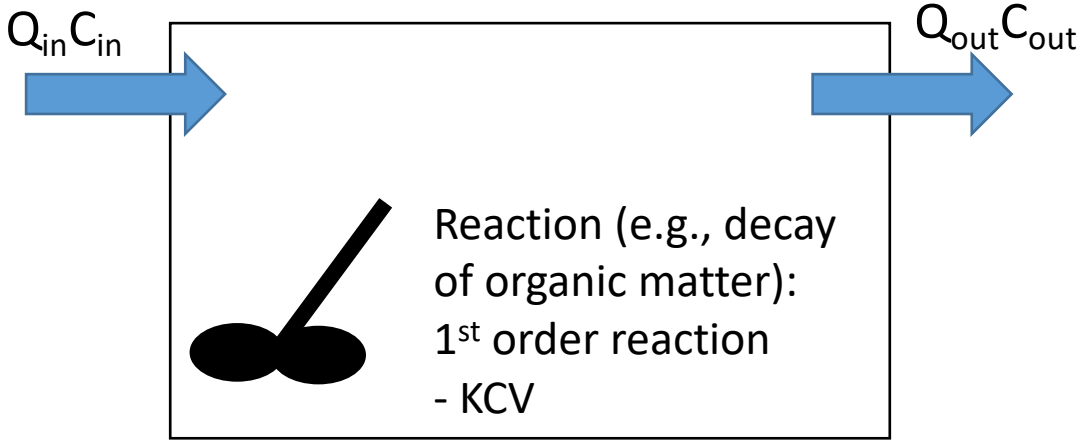
$Q_{in}C_{in} - Q_{out}C_{out} - VkC$

$Q_{in} = Q_{out}$

$C_{out} = C$

$QC_{in} - QC - VkC \Rightarrow C = \frac{QC_{in}}{Q + kV}$

Continuously stirred tank reactor (**CSTR**), also known as **Complete-mix** reactor



Assume 1st Order Reaction

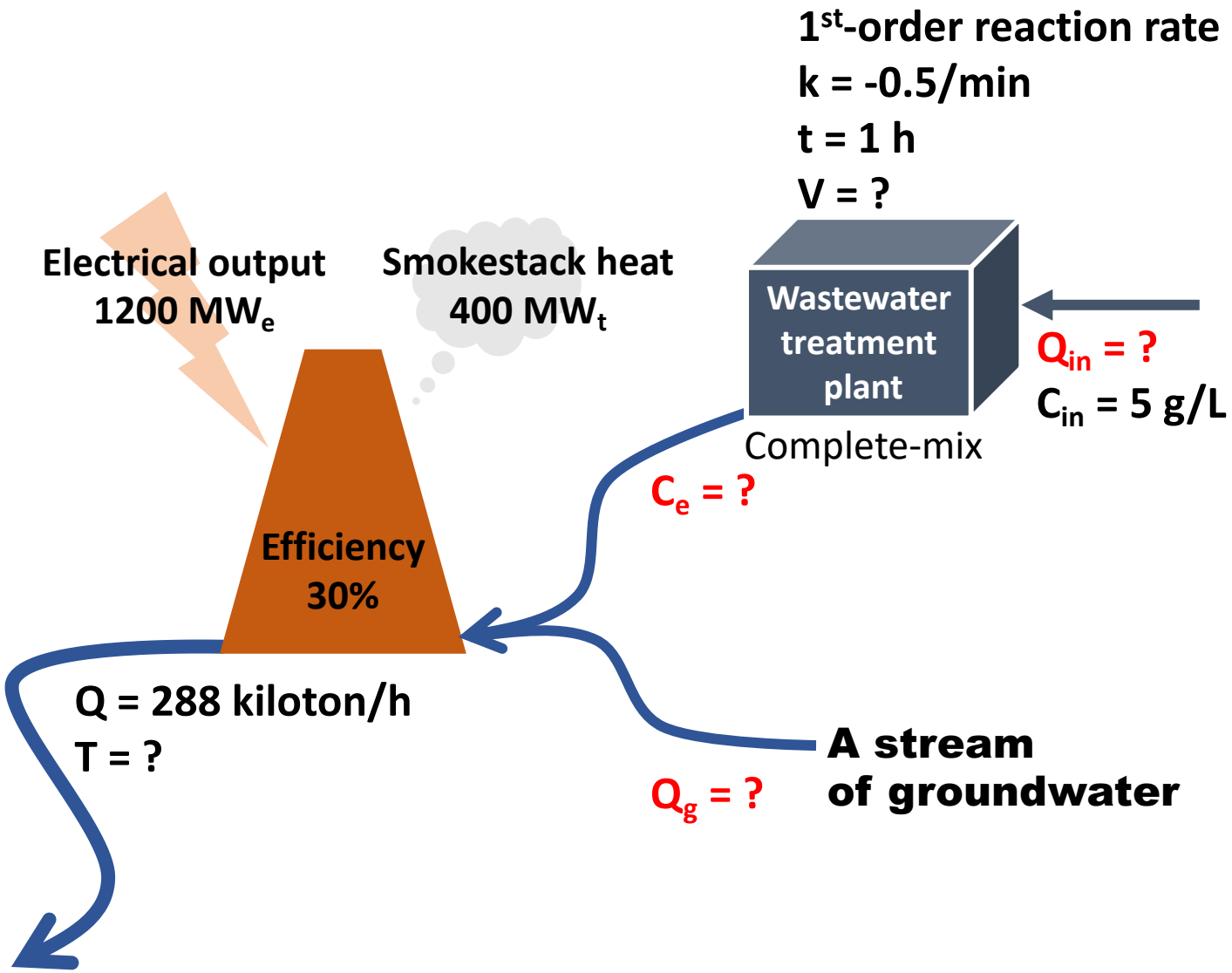
$$C_{out} = \frac{C_{in}}{1 + k\theta}$$

$\theta = \frac{V}{Q} = t_H$

Where θ is the hydraulic retention time (HRT), $\theta = V/Q$

$Q_{in}C_{in} - Q_{out}C_{out} - kC_{out}V = 0 \rightarrow C_{out} = QC_{in}/(Q + kV) = C_{in}/(1 + kV/Q)$

Past paper 2025-S1 Q11



1. What is the effluent concentration (C_e)? (2 marks)

$Q_{in}C_{in} - Q_eC_e - kC_eV = 0$
$C_e = C_{in}/(1+kt) = 5000\ ppm/(1 + 60*60*(0.5/60))$
$= 161\ ppm$

2. Design the volume (V) and input flow rate (Q_{in}) of the wastewater treatment plant and find out the flow rate Qg of groundwater stream. (4 marks)

Q_{in} 10 m^3/s
$Qg = Q - Qin = ((288*10^6 kg/(10^3 kg/m^3))/3600s)$
$- 10\ m^3/s = 70\ m^3/s$
$V = 60*60s*10 m^3/s = 36000\ m^3$

Only be provided in the present class

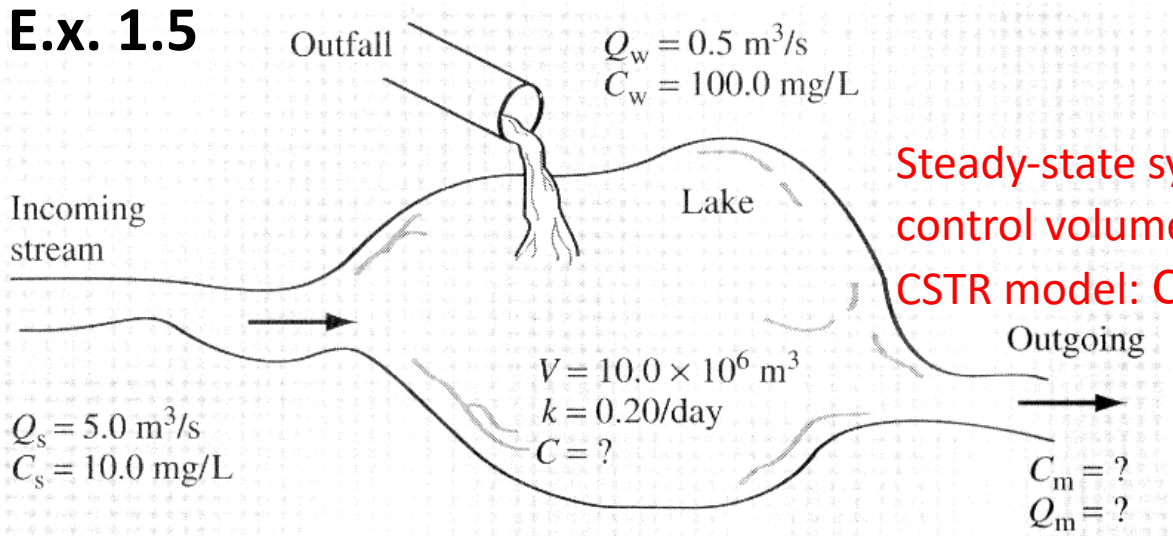
Mass balance

– Steady systems with nonconservative pollutants

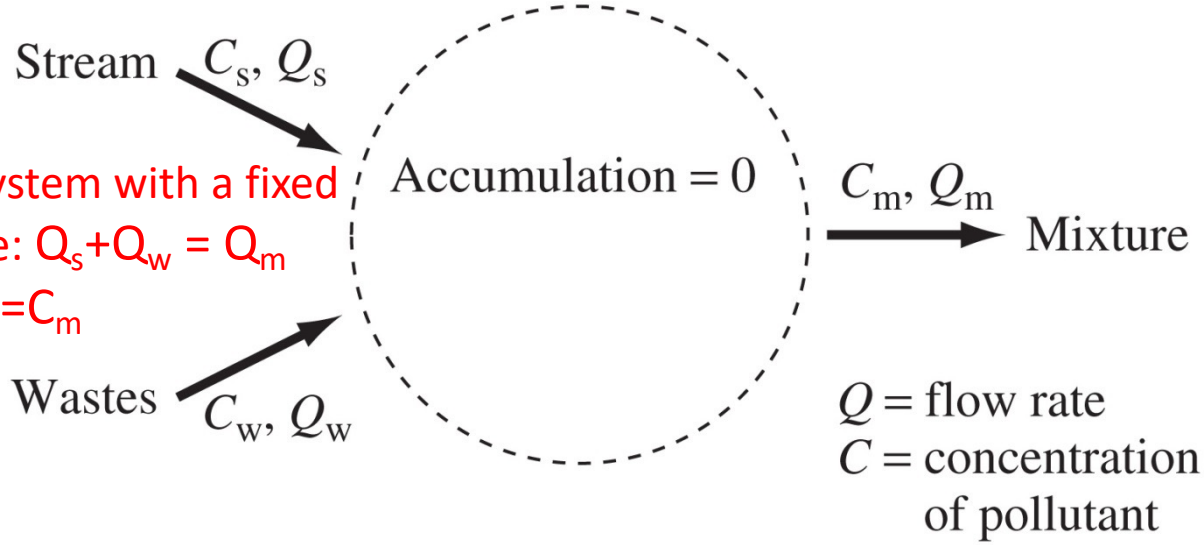
Reaction:
Decay reaction (-)
1st order → VkC
K = - 0.2/day

Continuously stirred tank reactor (**CSTR**)

E.x. 1.5



Steady-state system with a fixed control volume: $Q_s+Q_w = Q_m$
CSTR model: $C=C_m$



Textbook Fig 1.2

0 = input rate1 + Input rate2 – Output rate - reaction rate (decay)

$$0 = Q_sC_s + Q_wC_w - Q_mC - VkC$$

$$Q_mC + VkC = Q_sC_s + Q_wC_w$$

$$(Q_m + Vk)C = Q_sC_s + Q_wC_w$$

$$C = \frac{Q_sC_s + Q_wC_w}{Q_s+Q_w + Vk} = \frac{5\text{m}^3/\text{s}*10\text{g}/\text{m}^3 + 0.5\text{m}^3/\text{s}*100\text{g}/\text{m}^3}{(5.0 + 0.5)\ \text{m}^3/\text{s} + 0.2/\text{day} * 10*10^6\ \text{m}^3}$$

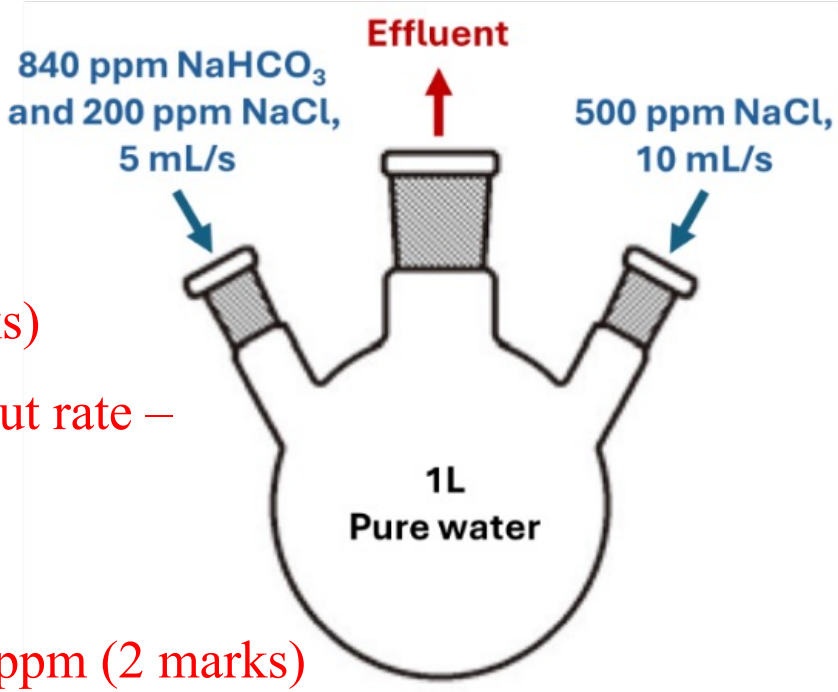
$$= 3.5\ \text{g}/\text{m}^3$$

Mass balance

– Steady systems with conservative pollutants

(Past paper) Consider an open three-neck flask containing 1 L of pure water (see diagram below). A water solution containing 840 ppm $NaHCO_3$ and 200 ppm NaCl is steadily poured into the flask with a flow rate of 5 mL/s from the left neck. Another water solution of 500 ppm NaCl is steadily input from the right neck with a flow rate of 10 mL/s. From the middle neck, an effluent is steadily pumped out of the flask. Assuming a steady-state system, the solution in the flask is completely mixed at standard temperature and pressure with 300 ppm CO_2 in the air, no water evaporation, and no decay/generation reaction in the system.

? Perform the mass balance for **water** and **NaCl** in the flask and find the steady-state concentration of NaCl in the effluent. **(4 marks)**



Mass balance for water: input rate1 + input rate2 = output rate

$\rho Q_1 + \rho Q_2 = \rho Q_{out} \rightarrow Q_1 + Q_2 = Q_{out}$ 5 mL+10 mL=15 mL (2 marks)

Mass balance for NaCl: accumulation rate = input rate1 + input rate2 – output rate – reaction rate

$0 = Q_1C_1 + Q_2C_2 - Q_{out}C_{out}$

$0 = 5\ mL/s \times 200\ ppm + 10\ mL/s \times 500\ ppm - 15\ mL/s \times C \rightarrow C = 400\ ppm$ (2 marks)

Ideal reactors

- Plug flow reactor (PFR)



Assume 0th Order Reaction

$$C_{out} = C_{in} - k\theta$$

Long pipe (Length >> Width)
No mixing of pollutants along the length

Assume 1st Order Reaction

$$C_{out} = C_{in}e^{-k\theta}$$

Distance x
Time = distance / velocity
Time = volume / flow rate t=A*x/Q
(A is cross-sectional area of the pipe)

For **each control volume**, model as a batch reactor!!!

Mass balance

$$Qc = \frac{\text{Volume}}{\text{time}} \times \frac{\text{mass}}{\text{Volume}} = \frac{\text{mass}}{\text{time}}$$

$$V\frac{dC}{dt} = QinCin - Q_{out}C_{out} + VkC^n$$

Accumulation rate = Input rate - Output rate + Reaction rate

Steady state → accumulation rate = 0

Conserved substance → reaction rate = 0

CSTR reactor → $C=C_{out}$

PFR reactor (batch system) →

Input rate = 0

Output rate = 0

Steps:

1. **Perform a generic mass balance equation** as above
2. **Define your system and model reactor**: steady-state? batch system? conserved? CSTR or PFR?
3. **Find out reaction type**: generation or decay? Zero or first order?

Mass balance

$$V\frac{dC}{dt} = QinCin - Q_{out}C_{out} + VkC^n$$

Accumulation rate = Input rate - Output rate + Reaction rate

Steady-state system → accumulation rate = 0 $\quad \mathbf{0 = Q_{in}C_{in} - Q_{out}C_{out} + VkC^n}$

Conserved system → reaction rate = 0 $\quad \mathbf{V\frac{dC}{dt} = Q_{in}C_{in} - Q_{out}C_{out}}$

Steady-state and conserved system → $\quad \mathbf{0 = Q_{in}C_{in} - Q_{out}C_{out}}$

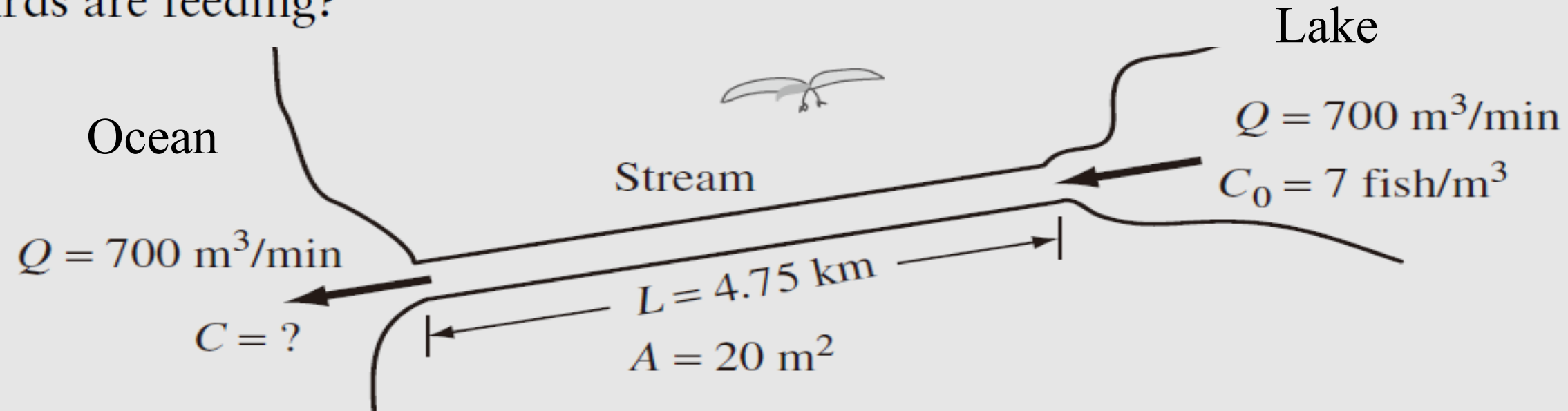
Two reactors:

CSTR reactor: Steady-state system → $\quad \mathbf{0 = Q_{in}C_{in} - Q_{out}C_{out} + VkC^n}$

PFR reactor: Batch system → input rate = 0 and output rate = 0 $\quad \mathbf{V\frac{dC}{dt} = VkC^n}$

Every year, herons, seagulls, eagles, and other birds mass along a 4.75 km stretch of stream connecting a lake to the ocean to catch the fingerling salmon as they migrate downstream to the sea. The birds are efficient fishermen and will consume 10,000 fingerlings per kilometer of stream each hour regardless of the number of the salmon in the stream. In other words, there are enough salmon; the birds are only limited by how fast they can catch and eat the fish. The stream's average cross-sectional area is 20 m^2, and the salmon move downstream with the stream's flow rate of 700 m^3/min. If there are 7 fingerlings per m^3 in the water entering the stream, what is the concentration of salmon that reach the ocean when the birds are feeding?

Zero-order reaction



Plug flow reactor (PFR) (Batch system)

Zero-order decay reaction: $VdC/dt=Vk \rightarrow C=C_0-kt$, so we don't have to care about the specific value of V

$k=10{,}000fish/(20m^2*1km)/hr = 10{,}000fish/(20m^2*1000m)/1hr = 0.5fish*m^{-3}*hr^{-1}$.

$t=V/Q = 4.75km*20m^2/(700m^3/min) = (4.75km*1000)m*20m^2/(700m^3/(min/60)h) = 2.26hr$

$C= 7fish/m^3 – 0.5\ fish/(m^3*hr) * 2.26hr = 5.9\ fish/m^3$

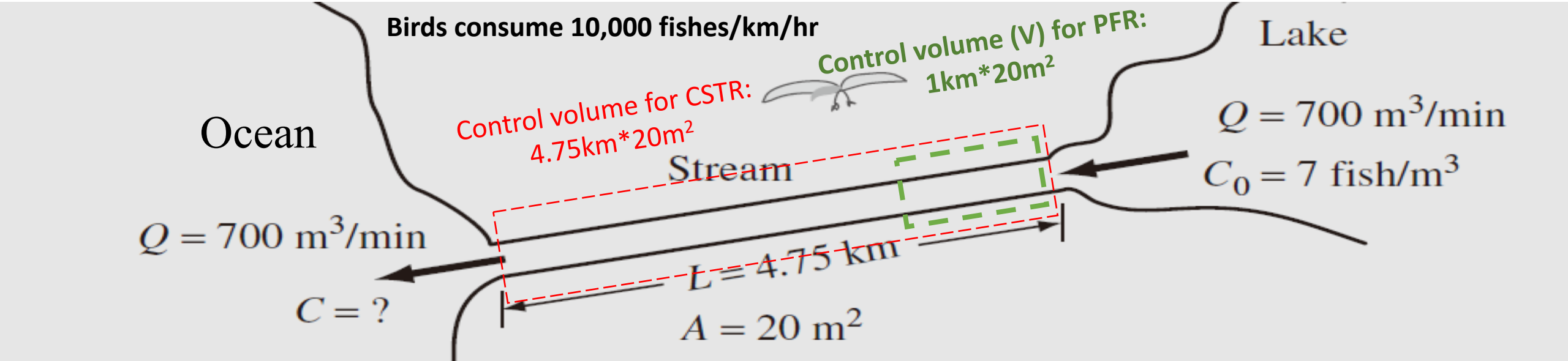
*Continuously stirred tank reactor (CSTR),

But the question does not give assumption of complete-mix system.

zero-order decay reaction: Vk

$0 = Q_{in}C_{in} - Q_{out}C_{out} – Vk = QC_0 + QC – Vk \rightarrow C = (QC_0 – Vk)/Q$

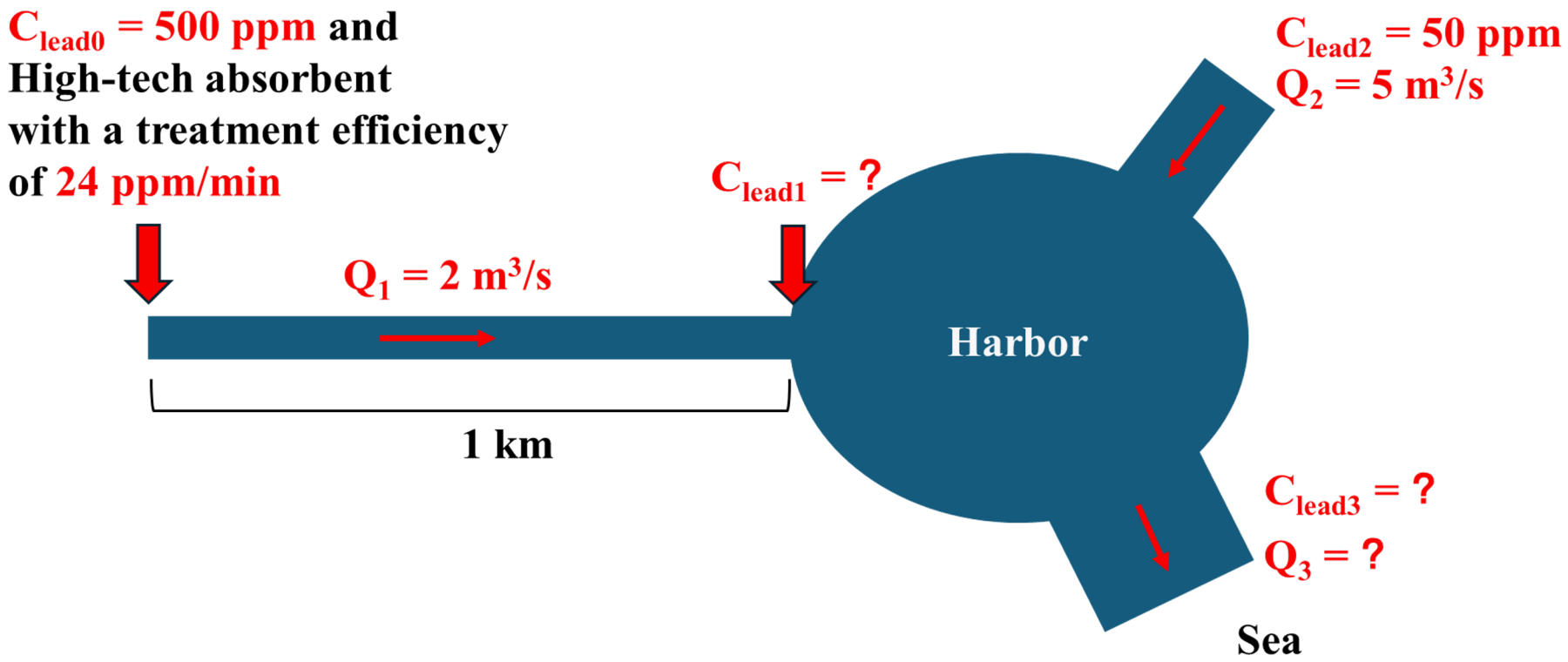
$C = [700m^3/min*7fish/m^3 – 4.75km*20m^2*10{,}000/(20m^2*1km)/60min]/(700m^3/min) = 5.9\ fish/m^3$



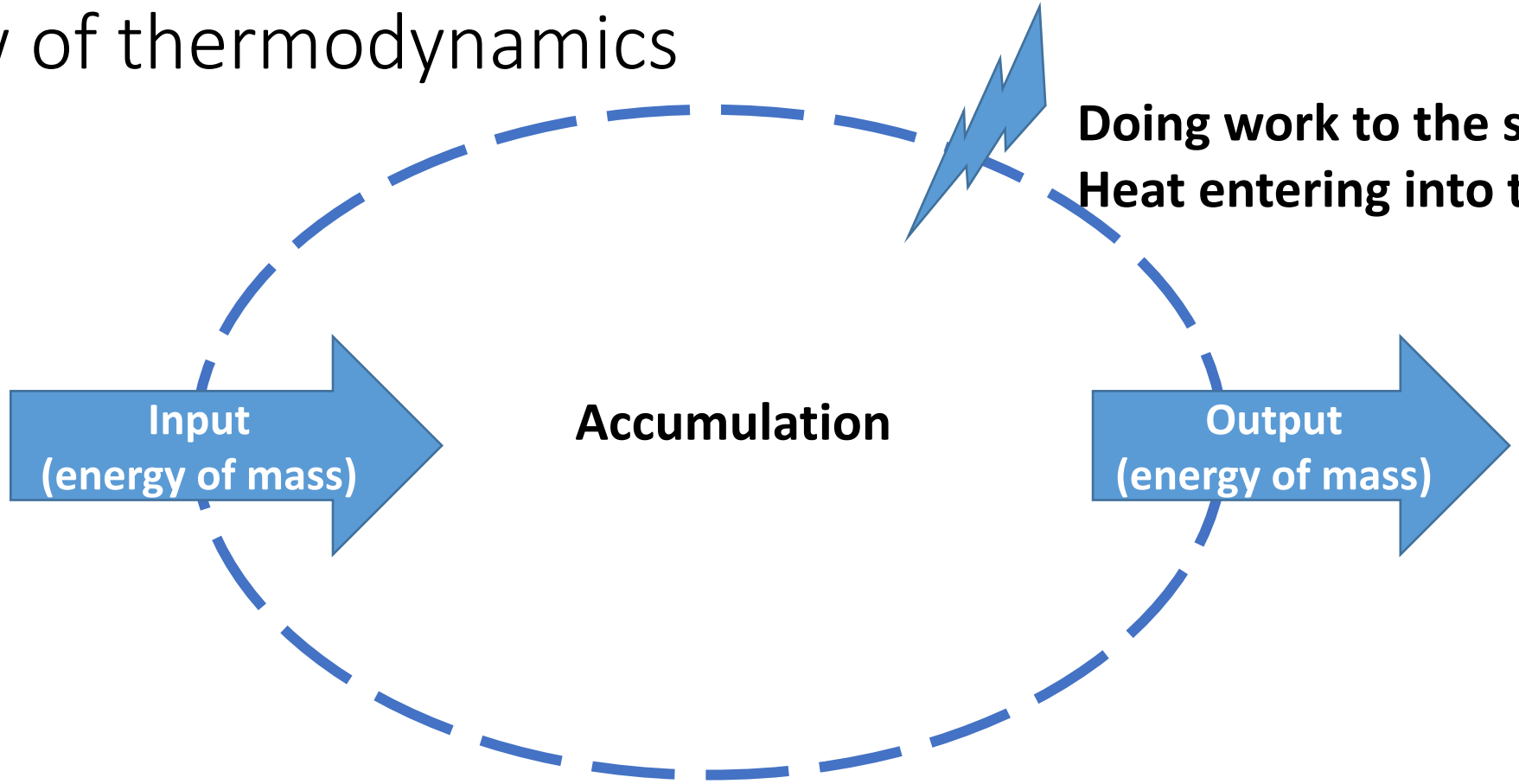
A wastewater stream carrying **500 ppm of lead (Pb)** is discharged into a harbor through a **1 km** long narrow tube. The cross-sectional area of the tube is **2 m^2**, and the discharge rate is **2 m^3/s** (see diagram below). To treat the heavy metals, some high-tech materials are added to the wastewater and flow along the tube to adsorb the lead at a rate of **24 ppm/min**. These materials are collected and recycled at the end of the tube (i.e., the materials will not be discharged into the harbor). In addition, a river containing **50 ppm** of lead flows into the harbor at a rate of **5 m^3/s**. The water in the harbor finally runs to the sea. Assuming a steady-state system at **room temperature and standard pressure**, the adsorption performance of high-tech materials is stable, and the harbor is completely mixed.

(1) C_{lead1} at the end of the tube?

(2) C_{lead3} discharged into the sea?



Energy fundamentals
- First Law of thermodynamics



Net change of energy In the system = **Input (energy of mass)** - **Output (energy of mass)** + **Work and heat**

Energy fundamentals
- Specific heat

$\Delta Q = cm\Delta T$ *(Q stands for energy)*

- Specific heat: the amount of heat required to raise the temperature of a unit mass by 1 degree
  - Specific heat of water c: 1 cal/g.°C = 1 kcal/kg.°C (Note: 1 Cal = 4.18 J)
- Specific heat for gas:
  - Specific heat at constant volume c_v (fixed volume)
  - Specific heat at constant pressure c_p (fixed pressure)
- Internal energy and enthalpy
  - ~~$\Delta U = m\, c_v \Delta T$~~
  - ~~$\Delta H = m\, c_p \Delta T$~~
  - ~~$H = U + PV$~~

$PV = nRT$

Energy fundamentals - Phase change

The energy required to cause a phase change of a unit mass at the same pressure is called the latent heat.

- When a substance changes phase, energy is absorbed or released without a change in temperature
  - Freeze/melt (latent heat of fusion)
  - Evaporate/condense (latent heat of vaporization)
  - Energy released or absorbed in phase change = *m*L, where L is the latent heat

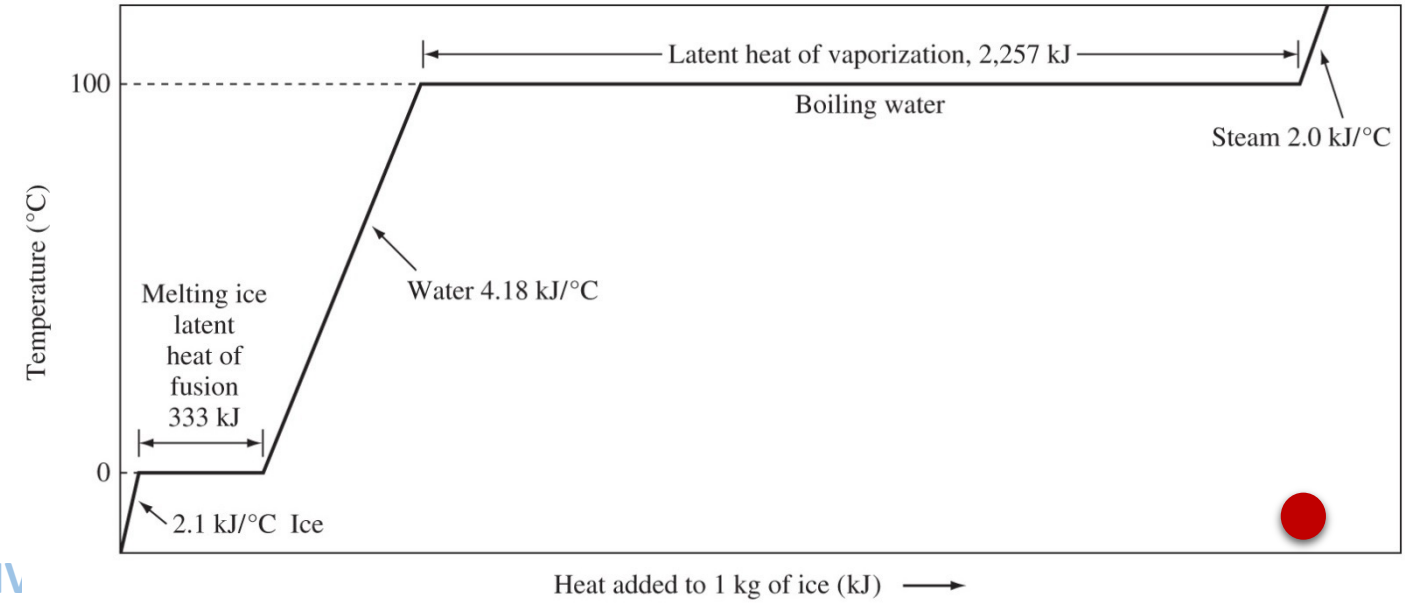


Figure: 01-13

Important properties of water	
Specific heat	4.18 kJ/kg.°C
Heat of vaporization (100 °C)	2257 kJ/kg
Heat of vaporization (15 °C)	2465 kJ/kg
Heat of fusion	333 kJ/kg
Density (at 4 °C)	1000 kg/m^3

Energy fundamentals - Phase change

Ex. 1.11 Power for the hydrologic cycle

1m/yr * 5.10 x 10^{14} m^2 —> volume/year

Global rainfall has been estimated to average about 1 m of water per year across the entire **5.10 x 10^{14} m^2** of Earth's surface. **Find the energy required to cause that much water to evaporate each year.** Compare this to the estimated 2007 world energy consumption of 4.7 x 10^{17} kJ/yr and compare it to the average rate at which sunlight strikes the surface of the Earth, which is about 168 W/m^2.

Solution:

The energy required to evaporate 1 kg of 15 °C water (roughly the average global temperature) is:

2465 kJ

Important properties of water	
Specific heat	4.18 kJ/kg.°C
Heat of vaporization (100 °C)	2257 kJ/kg
Heat of vaporization (15 °C)	**2465 kJ/kg**
Heat of fusion	333 kJ/kg
Density (at 4 °C)	1000 kg/m^3

Energy fundamentals - Phase change

Solution:

The total energy required to vaporize all of that water is:

1 m/yr × 5.10 x 10^{14} m^2 × 10^3 kg/m^3 × 2465 kJ/kg = **1.25 × 10^{21} kJ/yr**

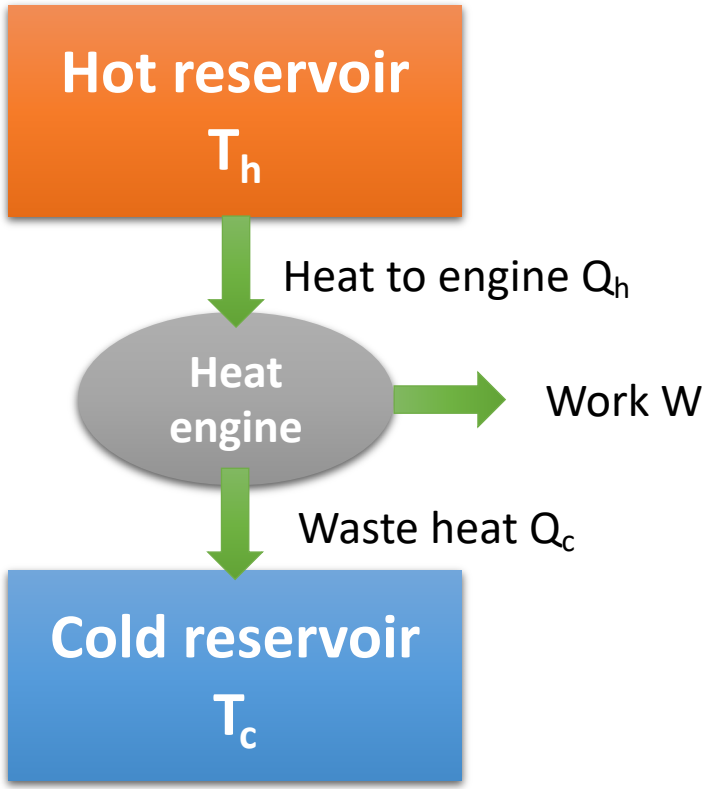
This is roughly 2700 times the 4.7 x 10^{17} kJ/yr of energy we use to power our society.

Averaged over the globe, the energy required to power the hydrologic cycle is

$$\frac{1.25\times10^{24}\ \mathrm{J/yr}\times1\ \frac{\mathrm{W}}{\mathrm{J/s}}}{365\ \mathrm{day/yr}\times24\mathrm{h/day}\times3600\mathrm{s/h}\times5.10\times10^{14}\mathrm{m}^2}=\mathbf{78.0}\ \mathrm{W/m}^2$$

which is equivalent to almost half of the 168 W/m^2 of incoming sunlight striking the Earth's surface.

Energy fundamentals
- Second Law of thermodynamics



Second law:

- ❑ Impossible to achieve 100% efficiency
- ❑ Loss in terms of waste heat

Efficiency

$$\eta = \frac{W}{Q_h}$$

Carnot efficiency (ideal efficiency)

$$\eta_{max} = 1 - \frac{T_c}{T_h}$$

Thermal power plant: Th ~ 600 °C; Tc ~ 20 °C

→ η_{max} = 1-(273+20)/(273+600) = 66%

Read Ex. 1.13

Energy fundamentals
- Phase change

Ex. 1.12

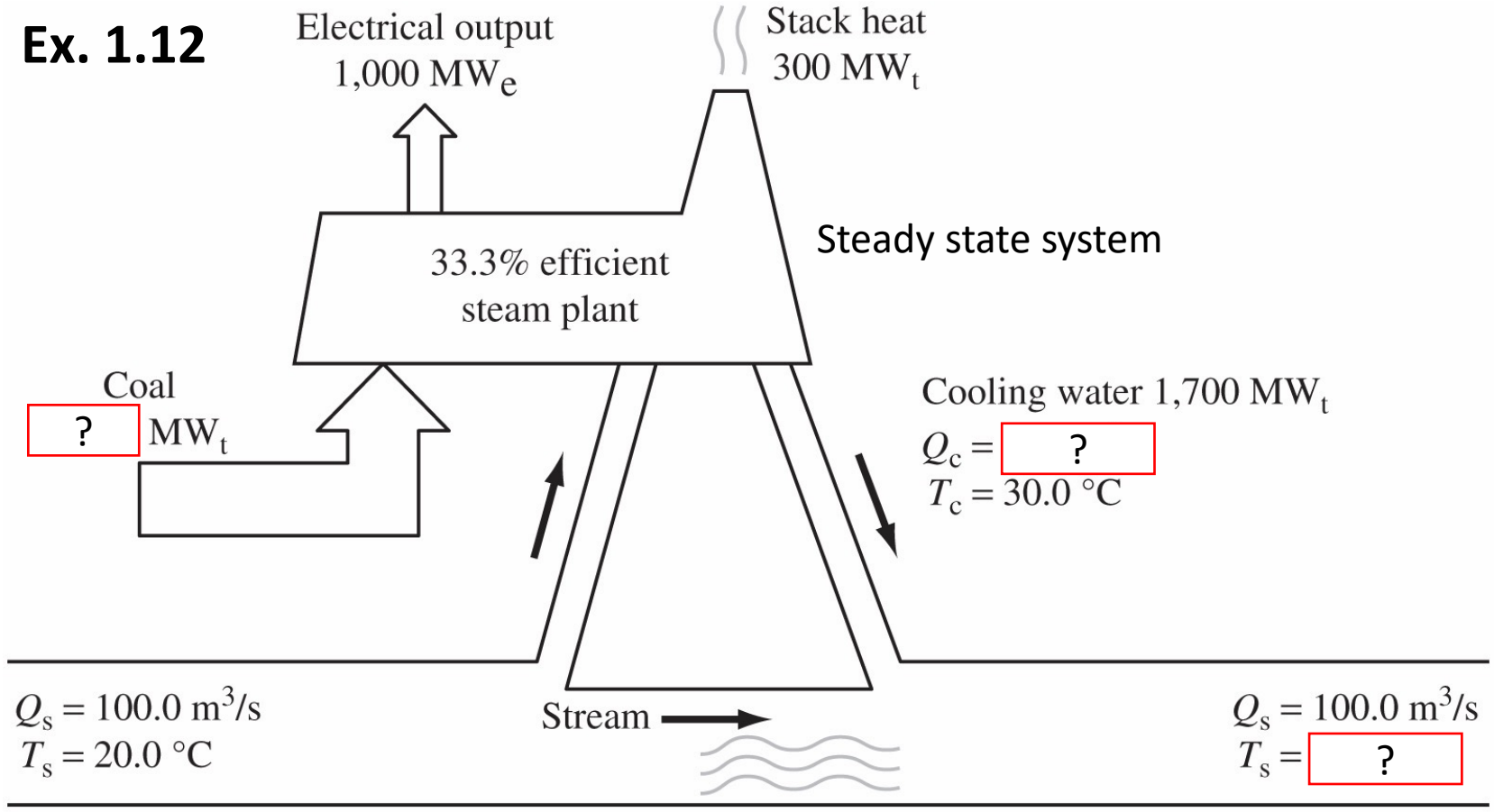


Figure: 01-14

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Readings and practice questions

- Readings
  - Essential reading: Sec 1.3, 1.4

- Practice questions
  - Q1.1, 1.3, 1.5, 1.11, 1.16, 1.29, 1.33